Accessible Torque Bandwidth of a Series Elastic Actuator Considering the Thermodynamic Limitations

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Motivation

- Neglecting thermal limitation analysis in torque control applications has a risk of causing irreversible internal demagnetisation damages.

- Torque control algorithms and thermodynamic models are often treated as separate branches for an actuator.

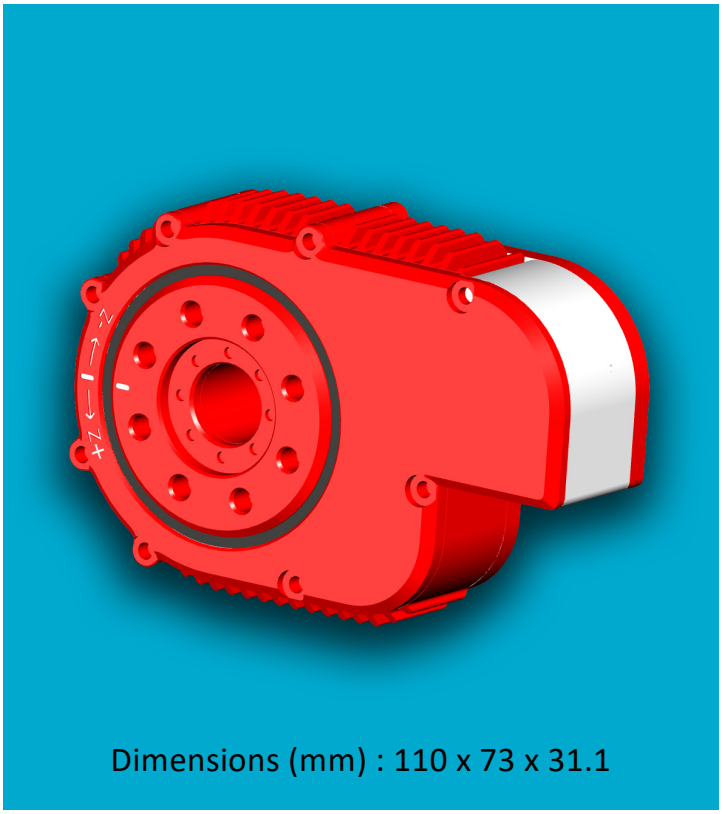
Content of the presentation

- System identification of a series elastic actuator
  - Electrical subsystem
  - Mechanical subsystem
  - Electromechanical control system
  - Thermodynamics
- Accessible torque bandwidth of a series elastic actuator
  - Open loop torque response
  - Closed loop torque tracking

Section 1:

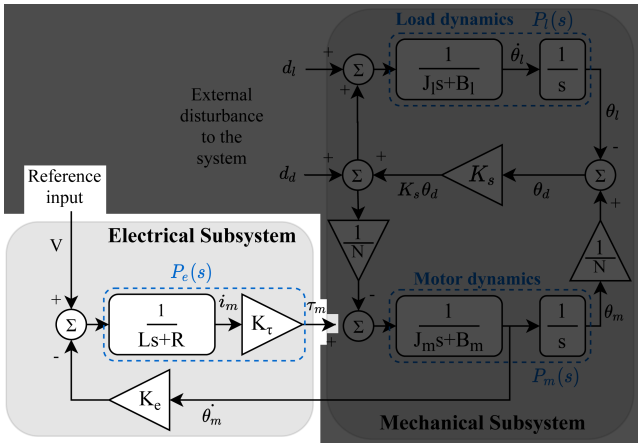
System Identification of a Series Elastic Actuator

HEBI Robotics X5-9 as a Case Study

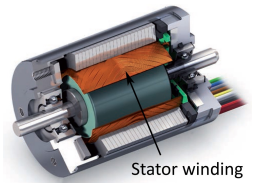


Dimensions (mm) : 110 x 73 x 31.1

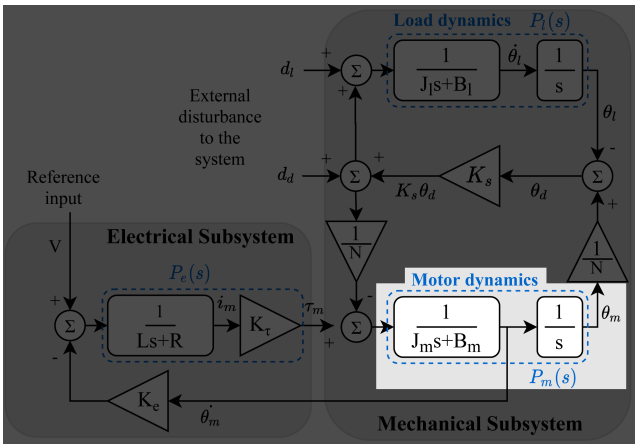
Electrical Subsystem



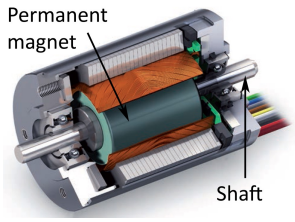
- V(t) : winding voltage
- K_e : back e.m.f. constant
- R : winding resistance
- L : winding inductance
- $i_m(t)$: winding current
- K_τ : motor torque constant
- $\tau_m(t)$: electromagnetic torque



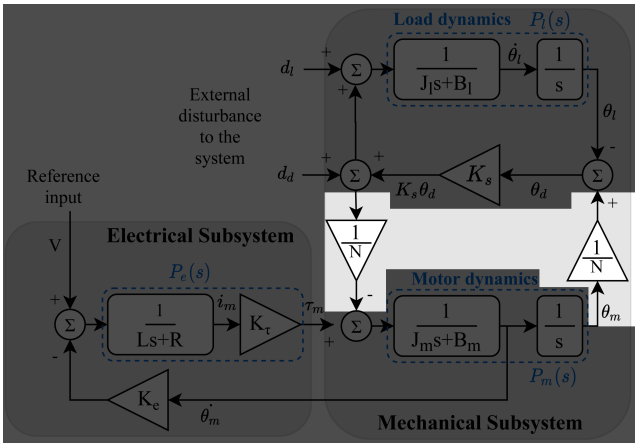
Mechanical Subsystem – Motor Dynamics



- J_m : rotor inertia
- B_m : viscous friction
- θ_m : motor angle

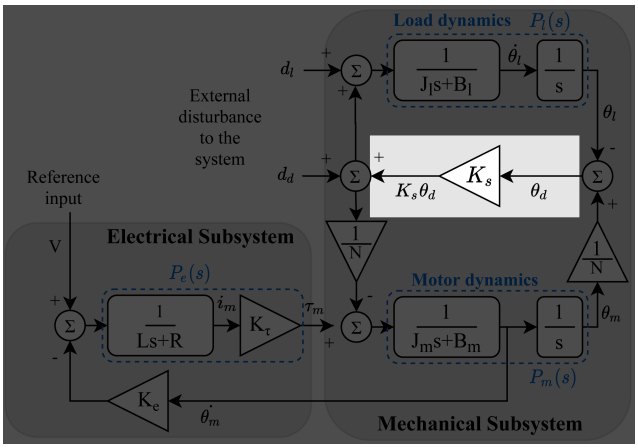


Mechanical Subsystem – Gear Reduction

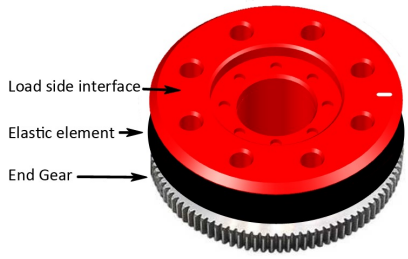


- θ_m : motor angle
- N : gear ratio

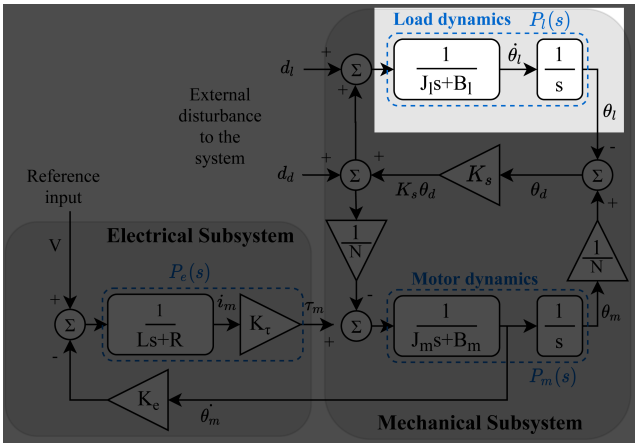
Mechanical Subsystem – Elastic Element



- K_s : elastic constant
- θ_d : deflection of the elastic element

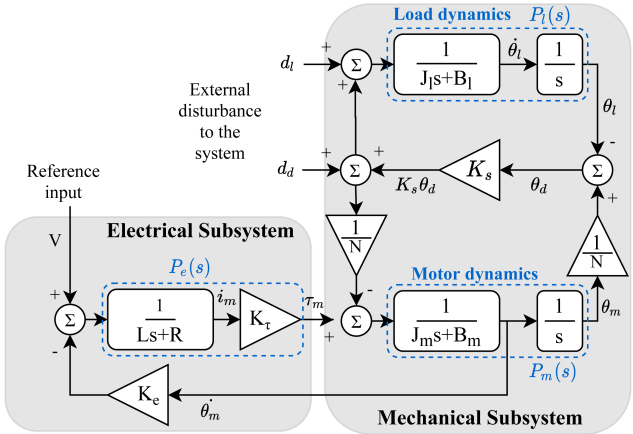


Mechanical Subsystem – Load Dynamics



- J_l : load inertia
- B_l : damping coefficient
- θ_l : load angle

Electromechanical Subsystem

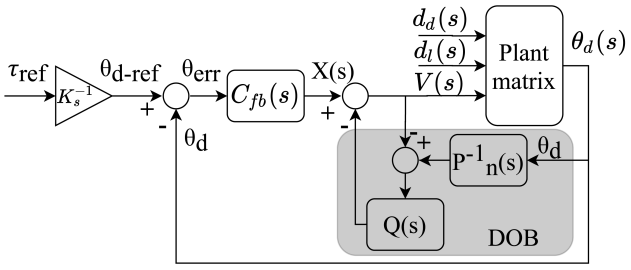


- d_l : external disturbances coming into the load dynamics
- d_d: external disturbances coming into the motor dynamics

Electromechanical Subsystem

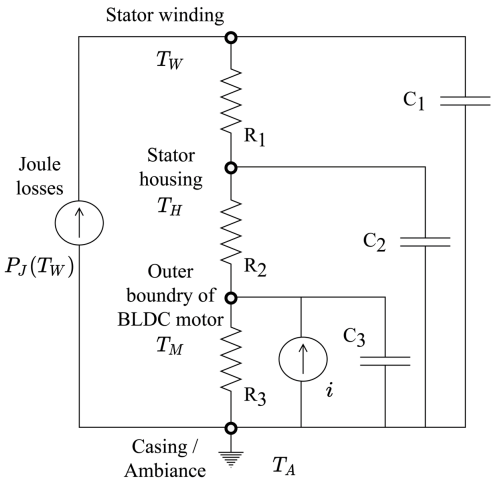
- Input: an open loop frequency sweep signal
- Output : fixed; simulating an infinite load side inertia ($J_l \rightarrow \infty$).
- The system transfer function reduces to;
- $$\frac{\theta_d(s)}{V(s)} = \frac{\frac{K_\tau}{N}}{J_m L s^3 + (B_m L + J_m R)s^2 + \left(K_e K_\tau + \frac{K_S L}{N^2} + B_m R\right)s + \frac{K_S R}{N^2}}$$

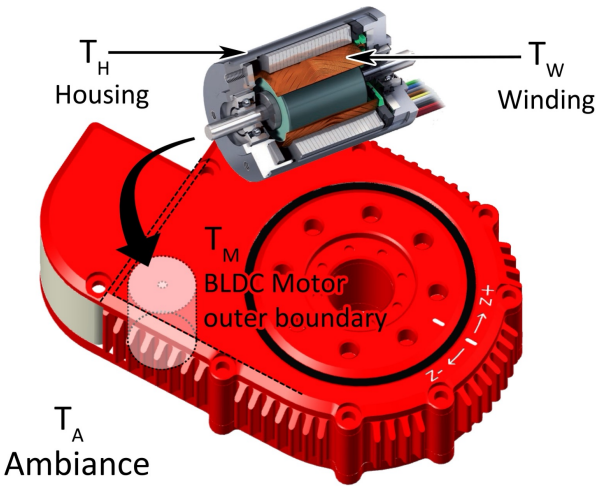
Electromechanical Control System



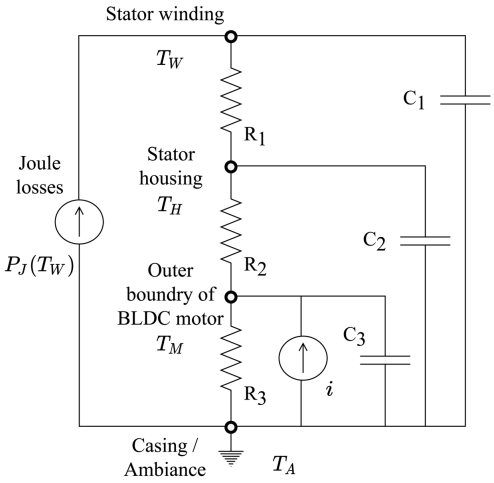
- d_d and d_l : external disturbances
- θ_d : deflection of the elastic element
- K_s : elastic constant
- Q(s) : Butterworth low pass filter
- $P_n(s)$: nominal plant model
- $C_{fb}(s)$: feedback controller
- DOB : disturbance observer

Thermodynamics



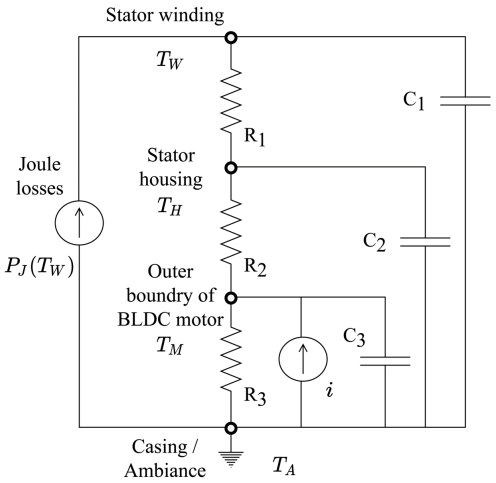


Thermodynamics



- Thermal resistances R_1 to R_3 characterise the heat dissipation between different bodies.
- C_1, C_2 and C_3 represent the heat capacity constants.
- Resultant power flow from solid-state electronic heat flux is modelled using a current source i.

Thermodynamics



- Continuous operation
  - Temperatures reach some steady state values.
- Periodic operation
  - Temperatures reach some steady state values according to the periodic operation.
- Short-term overloading
  - Maximum ON time is defined.
- Arbitrary operation

Thermodynamics – Continuous Operation

Playback x1

- At the steady state condition, the capacitors are saturated and can be open circuited.
- Steady state results as well as curve fitting results can be used to calculate the unknown constants of the model.
- Legend: **W**:Winding, **H**:Housing, **M**:Motor boundary, **B**:Circuit board, **A**: Ambiance

Thermodynamics – Arbitrary Operation

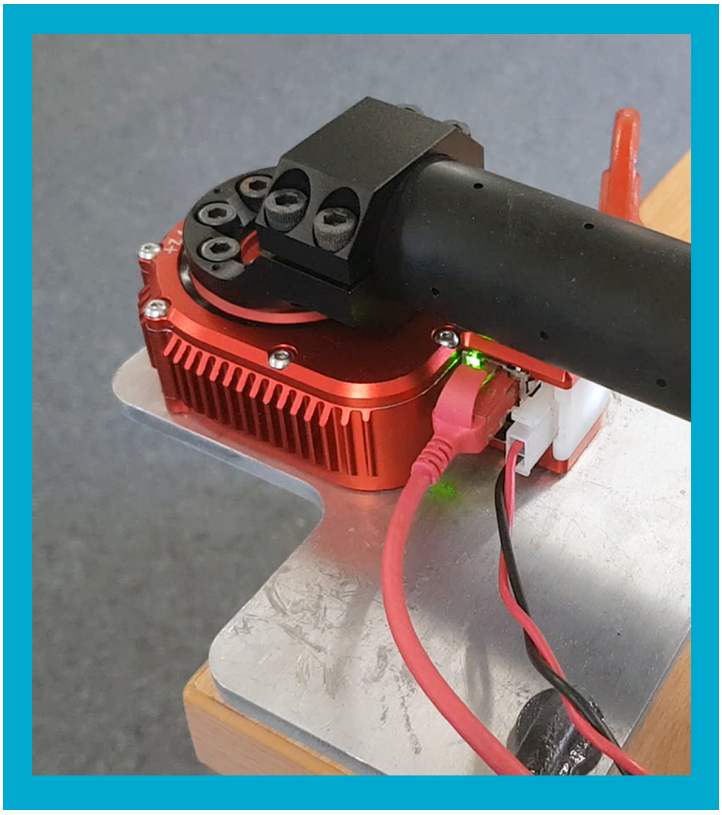
- Simplifying the thermodynamic model, it is assumed that the ambient temperature is equal to the temperature just outside the BLDC motor and $\tau_1 << \tau_2$.

- $T_W(t) = T_H(t) - T_W(0)e^{\frac{-1}{R_1 C_1}t} + \int_0^t P_J(\tau)g(t-\tau)d$
- $g(t) = \frac{1}{C_1} e^{-\frac{1}{R_1.C_1}t}$

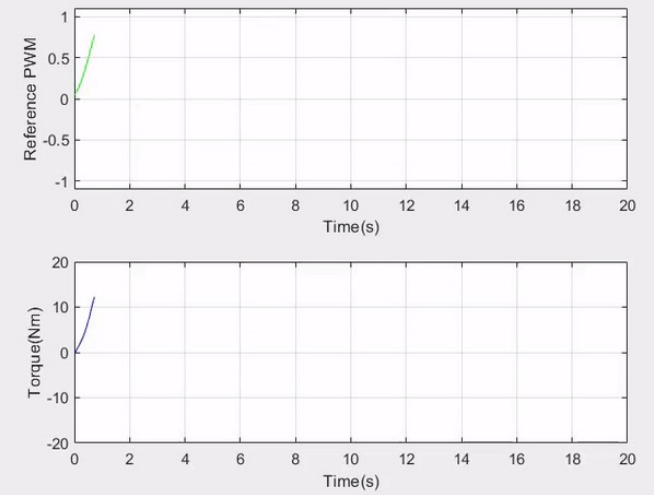
Section 2:

Accessible Torque Bandwidth

Considering the Thermodynamic Limitations

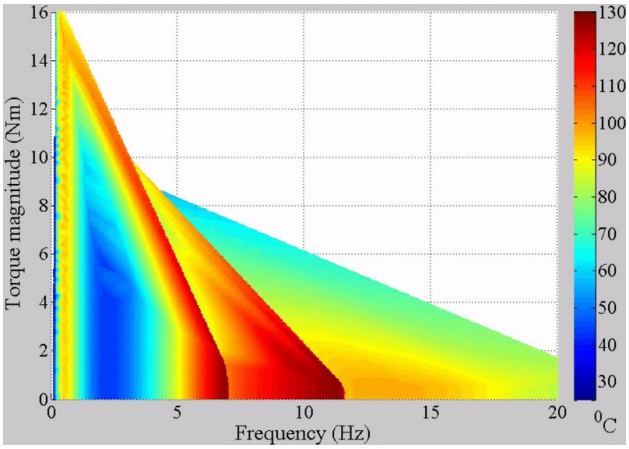


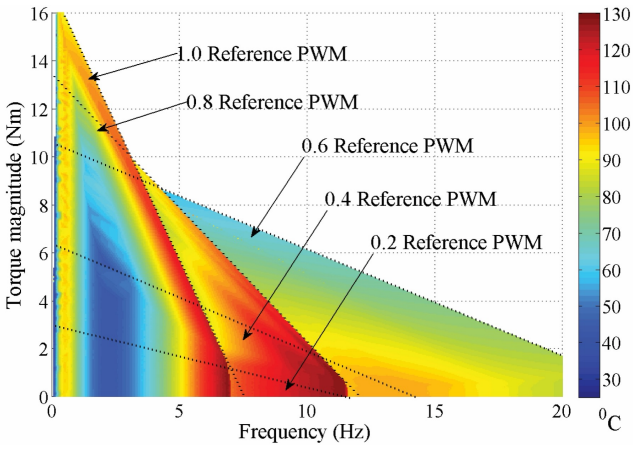
Open Loop Torque Response



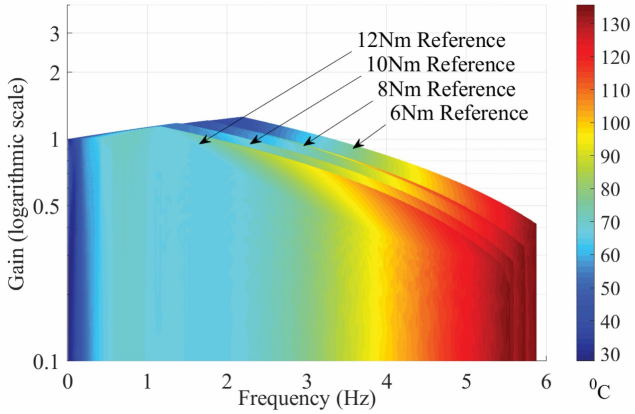
- The torque is effectively generated according to the open loop plant dynamics.
- Fixed-output SEA is commanded with a sinusoidal reference PWM signal which linearly increases in the frequency

Open Loop Torque Response



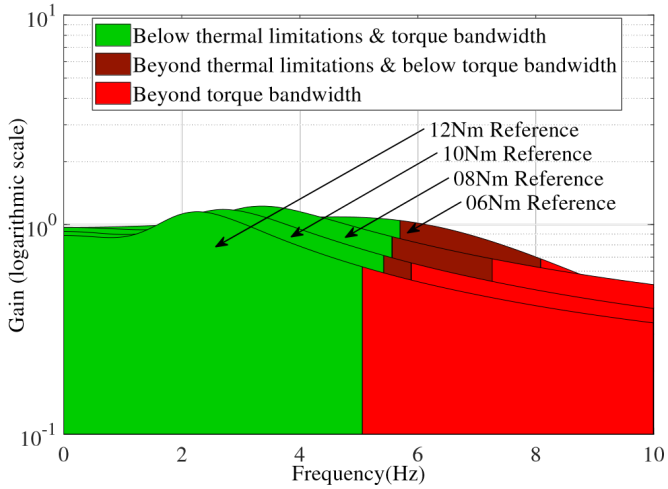


Closed Loop Torque Tracking



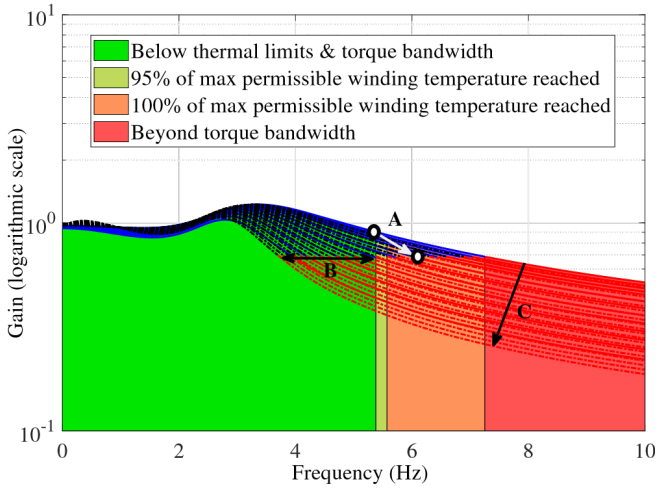
- Characteristics are governed by the feedback controller.
- Thermal behaviour is similar as the feedback controller drives the input to saturation regions at higher frequencies.

Closed Loop Transfer Functions



- For 06, 08 and 10 Nm reference values, the actuator could not reach the characteristic closed loop torque bandwidth due to the occurrence of thermal extremes in a prior frequency value.

Improving Accessible Torque Bandwidth



- Thermal controller is triggered at the threshold value of 95% of the maximum permissible winding temperature.
- Non-linear filtering and input reference signal amplitude restriction to lower the winding temperature.

Discussion and Conclusion

- Electromechanical torque bandwidth of 7.25 Hz was not accessible due to the stator winding temperature exceeding the maximum permissible value at 5.57 Hz.
- Inclusion of the thermal controller improves the accessible torque bandwidth by 8.25% up to 6.03 Hz.
- Inclusion of thermal controllers maximise the lifespan of actuators in torque control applications.

Thank you

